CHAPTER EXERCISE: Answers for this chapter start on page 320.

A calculator should NOT be used on the following questions.

1

If $f(x) = -2x^2 - 3x + 1$, what is the value of $|f(1)|$?

A) 3
B) 4
C) 5
D) 6

2

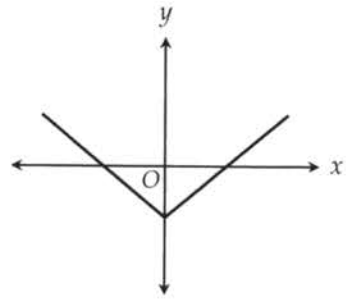
If $|2 - x| > 5$ and x is a positive integer, what is the minimum possible value of x ?

3

Which of the following expressions is equal to -5 for some value of x ?

A) $|x - 6| + 2$
B) $|x - 2| - 6$
C) $|x + 2| + 6$
D) $|x + 6| - 2$

4



Which of the following could be the equation of the function graphed in the xy-plane above?

A) $y = -|x| - 2$
B) $y = |x| - 2$
C) $y = |x| + 2$
D) $y = |x - 2|$

5

If $|x - 3| > 10$, which of the following could be the value of $|x|$?

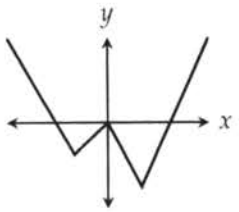
A) 2
B) 4
C) 6
D) 8

A calculator is allowed on the following questions.

6

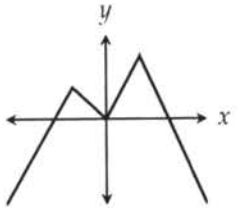
How many different integer values of x satisfy $|x + 6| < 3$?

7

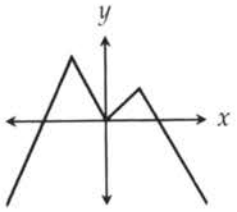


The graph of the function f is shown in the xy-plane above. Which of the following could be the graph of the function $y = |f(x)|$?

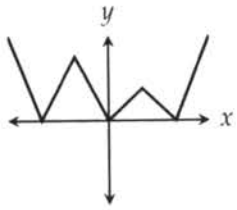
A)



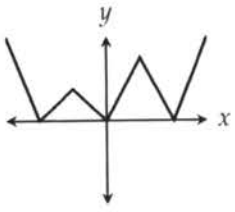
B)



C)



D)



8

If $|n - 2| = 10$, what is the sum of the two possible values of n ?

A) 4

B) 6

C) 12

D) 20

9

If $|x - 10| = b$, where $x < 10$, then which of the following is equivalent to $b - x$?

A) -10

B) 10

C) $2b - 10$

D) $10 - 2b$

10

A hot dog factory must ensure that its hot dogs are between $6\frac{1}{4}$ inches and $6\frac{3}{4}$ inches in length. If h is the length of a hot dog from this factory, then which of the following inequalities correctly expresses the accepted values of h ?

A) $\left|h - 6\frac{1}{4}\right| < \frac{1}{4}$

B) $\left|h - 6\frac{1}{4}\right| < \frac{1}{2}$

C) $\left|h - 6\frac{1}{2}\right| < \frac{1}{4}$

D) $\left|h - 6\frac{1}{2}\right| > \frac{1}{4}$

11

$$|n-2|<5$$

How many integers n satisfy the inequality above?

A) Six
B) Seven
C) Eight
D) Nine

12

Rolls of tape must be made to a certain length. They must contain enough tape to cover between 400 feet and 410 feet. If l is the length of a roll of tape that meets this requirement, which of the following inequalities expresses the possible values of l ?

A) $|l-400|<10$
B) $|l-405|>5$
C) $|l+405|<5$
D) $|l-405|<5$

13

If $|4x-4|=8$ and $|5y+10|=15$, what is the smallest possible value of xy ?

A) -20
B) -15
C) -5
D) -1

14

If $|a|<1$, then which of the following must be true?

I. $\frac{1}{a}>1$
II. $a^2<1$
III. $a>-1$

A) III only
B) I and II only
C) II and III only
D) I, II, and III

15

A bakery standardizes muffins to weigh between $1\frac{3}{4}$ and $2\frac{1}{4}$ ounces. If m is the weight of a muffin from this bakery, which of the following inequalities expresses the possible values of m ?

A) $\left|m-1\frac{3}{4}\right|<\frac{1}{4}$
B) $|m-2|<\frac{1}{4}$
C) $|m-2|<\frac{1}{2}$
D) $\left|m-1\frac{3}{4}\right|<\frac{1}{2}$

Chapter 20: Absolute Value

CHAPTER EXERCISE:

1. $\boxed{B}$

$$|f(1)| = |-2(1)^2 - 3(1) + 1| = |-4| = 4$$

2. $\boxed{8}$ The best way to solve this question is trial and error. If $x = 3$, for example, $|2 - 3| = 1$, which is not greater than 5. This result indicates that we should try larger numbers. If we continue to work our way up, we would arrive at the minimum possible value $x = 8$, which results in $|2 - 8| = 6$.

3. $\boxed{B}$ Only the expression in answer (B) can equal -5 (when $x = 1$ or 3). Because the absolute value of anything is always greater than or equal to 0, the other answer choices can never reach -5.

4. $\boxed{B}$ Recall that the graph of $y = |x|$ is a V-shape centered at the origin. The graph pictured is also V-shaped but converges at $y = -2$, which means it has shifted two units down. Therefore, the equation of the graph is $y = |x| - 2$. Note that $y = |x - 2|$ shifts the graph two units to the right, NOT two units down.

5. $\boxed{D}$ Test each of the answer choices, making sure to include the negative possibilities. For example, the answer is not (A) because when $x = 2$ or -2, $|x - 3|$ is not greater than 10. However, $|x - 3|$ is greater than 10 when $x = -8$.

6. $\boxed{5}$ Smart trial and error is the fastest way to find the bounds for x. The lower bound for x is -8 and the upper bound is -4. There are 5 integers between -4 and -8 (inclusive). If we wanted to do this problem more mathematically, we could set up the following equation:

$$-3 < x + 6 < 3$$

Subtracting 6,

$$-9 < x < -3$$

Since x is an integer,

$$-8 \leq x \leq -4$$

7. $\boxed{D}$ In the graph of $|f(x)|$, all points with negative y-values (below the x-axis) are flipped across the x-axis. All points with positive y-values stay the same. Graph (D) is the one that shows this correctly.

8. $\boxed{A}$ If n is positive,

$$n - 2 = 10$$
$$n = 12$$

If n is negative,

$$n - 2 = -10$$
$$n = -8$$

The sum of these two possible values of n is $12 + (-8) = 4$.

9. [C] Make up a number for x. Let's say $x = 3$. Then $b = |3 - 10| = 7$, and $b - x = 7 - 3 = 4$. Using our numbers, we're looking for an answer choice that gives 4 when $b = 7$. The only one that does so is (C).

   To do this question mathematically, we have to realize that when $x < 10$, $x - 10$ is always negative. Therefore,

$$x - 10 = -b$$
$$x = 10 - b$$

   Using substitution, $b - x$ becomes $b - (10 - b) = 2b - 10$.

10. [C] The midpoint of $6\frac{1}{4}$ and $6\frac{3}{4}$ is the average: $\left(6\frac{1}{4} + 6\frac{3}{4}\right)/2 = 6\frac{1}{2}$. The midpoint is $\frac{1}{4}$ away from the boundaries of the accepted range for the length of a hot dog. So whatever h is, it must be within $\frac{1}{4}$ of the midpoint:

$$\left|h - 6\frac{1}{2}\right| < \frac{1}{4}$$

11. [D] Smart trial and error is the fastest way to find the bounds for n. The lower bound for n is -2 and the upper bound is 6. There are 9 integers between -2 and 6 (inclusive). If we wanted to do this problem more mathematically, we could set up the following equation:

$$-5 < n - 2 < 5$$

   Adding 2,

$$-3 < n < 7$$

   Since n is an integer,

$$-2 \leq n \leq 6$$

12. [D] The midpoint of 400 and 410 is the average: $(400 + 410)/2 = 405$. The midpoint is 5 away from the boundaries of the accepted range for the length of a roll of tape. So whatever l is, it must be within 5 of the midpoint:

$$|l - 405| < 5$$

13. [B] There are two possible values of x, 3 and -1. There are two possible values of y, 1 and -5. We get the smallest possible value of xy when $x = 3$ and $y = -5$, in which case $xy = -15$.

14. [C] If $|a| < 1$, then by definition,

$$-1 < a < 1$$

   This means that III is true. Because a must be a fraction, $a^2 < 1$, so II is also true. However, I is not always true because when a is negative, $\frac{1}{a}$ is not greater than 1.

15. [B] The midpoint of $1\frac{3}{4}$ and $2\frac{1}{4}$ is the average: $\left(1\frac{3}{4} + 2\frac{1}{4}\right)/2 = 2$. The midpoint is $\frac{1}{4}$ away from the boundaries of the accepted range for the weight of a muffin. So whatever m is, it must be within $\frac{1}{4}$ of the midpoint:

$$|m - 2| < \frac{1}{4}$$